

Is AI Labor-Augmenting? Evidence from Korean Firms*

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Abstract

Does Artificial Intelligence (AI) adoption bias technological change toward labor? We estimate the effect of AI adoption on labor-augmenting and Hicks-neutral productivity for Korean firms from 2017 to 2024. AI adoption raises labor-augmenting productivity by 6.1% and leaves Hicks-neutral productivity indistinguishable from zero, and the adoption effect on output is 1.6%. Manufacturing carries twice the labor-augmenting coefficient of information and communication (ICT). The labor efficiency comes through production processes in manufacturing and through product and service development in ICT. We recover firm markups and find no evidence that AI adoption raises them.

Keywords: Artificial Intelligence (AI); Labor-Augmenting Productivity; Technology Adoption; Biased Technological Change; Market Power; Korean Firms

JEL Codes: D24, O33, J24, L11

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1 Introduction

Artificial Intelligence (AI) is spreading through firms in every sector of the economy. In Korea, the share of firms reporting that they use AI increased from 1.3% in 2017 to 8.8% in 2024. Studies find productivity gains at the task level (Noy and Zhang, 2023; Brynjolfsson et al., 2025) and at the firm level, where the outcome is a total factor productivity that is Hicks-neutral (Czarnitzki et al., 2023; McElheran et al., 2025). AI is a general-purpose technology that workers use. Does the technical change it carries make labor more efficient, or is it Hicks-neutral, raising the productivity of every input alike?

We estimate the effect of AI adoption on both labor-augmenting and Hicks-neutral productivity for Korean firms from 2017 to 2024. The pooled results show that AI adoption increases labor-augmenting productivity by 6.1% and leaves Hicks-neutral productivity indistinguishable from zero, so the 1.6% effect on output runs mainly through labor efficiency. The labor-augmenting shifter is positive and significant both before and after generative models arrived, and for early and late adopters alike. Manufacturing’s labor-augmenting coefficient is twice that of information and communication (ICT). The labor-augmenting effect comes through production processes in manufacturing and through product and service development in ICT, while no area shifts Hicks-neutral productivity.

To recover the two productivity states separately, we estimate a translog production function with a labor-augmenting and a Hicks-neutral productivity state. Each follows its own controlled Markov process with AI adoption and industry-year fixed effects as its shifters. Identification comes from the firm’s cost minimization over labor and the intermediate input. Dividing the two first-order conditions removes the markup, output and its price, and writes labor-augmenting productivity as a function of observed quantities and cost shares. We estimate equations in quasi-differences by dynamic panel GMM (Arellano and Bond, 1991; Blundell and Bond, 2000).

The pooled estimates come from 78,442 firm-year observations on 15,829 firms. Labor and

the intermediate input are gross complements, at an elasticity of substitution of 0.59. Returns to scale are a little below one. Adoption significantly shifts labor-augmenting productivity by 6.1% and leaves Hicks-neutral productivity indistinguishable from zero. On the output scale, AI adoption is worth 1.4% through labor efficiency against 1.6% in total.¹

When we split adoption by calendar time and by when a firm first adopted, the labor-augmenting effect stays positive and significant in every cell, from 1.20% to 1.86% on the output scale. In the two years after generative models arrived, firms that already had AI and firms taking it up for the first time differ instead in the Hicks-neutral channel, with total effects of 3.47% and 0.08%. Those two cells cover the same years, so the comparison is between the two groups of adopters, and it cannot separate who they are from how recently they adopted.

Technology need not be common across sectors, and the estimator supports two of the sectors, manufacturing and ICT. Manufacturing is the biggest sector, with half of the firms in the sample, and ICT has the highest adoption rate, while the other sectors have too few firms that ever switch. Manufacturing carries twice the labor-augmenting coefficient of ICT. That coefficient applies to little more than half as much labor in manufacturing, so on the output scale the two are 0.98% and 0.87%, and both are distinguishable from zero.

Each adopter reports the main area in which it applies AI, and the four areas are mutually exclusive. The labor-augmenting effect differs across them in manufacturing and ICT, while the Hicks-neutral effect is zero everywhere. Production processes have a positive effect on labor-augmenting productivity in manufacturing, but not in ICT. Product and service development has a positive effect in ICT. These sector patterns carry into the pooled sample, where both areas have a positive effect.

Estimating the production function with labor-augmenting productivity by cost minimization also recovers a markup robust to the firm’s conduct ([Doraszelski and Jaumandreu, 2019](#); [Demirer, 2026](#)). We find no evidence that adoption raises it.

¹Labor-augmenting productivity increases output through labor, and Hicks-neutral productivity raises it directly, so we put the two on a common scale. Subsection [3.2](#) shows the comparison.

Aggregated over the sample, adoption raises 2024 output by 0.31%. Adopters are 8.8% of firms unweighted and 35.5% revenue-weighted. These are large firms, and labor’s share of variable cost falls with size, so the labor-augmenting gain at an adopter is 0.64% of its output against 1.4% at the sample mean.

This paper relates to three literatures. First, our estimates come out of the literature that measures factor-biased technological change in firm data. Recent studies recover factor-specific productivity from firm panels. [Doraszelski and Jaumandreu \(2018\)](#) estimate labor-augmenting and Hicks-neutral efficiency for Spanish manufacturing, [Raval \(2019\)](#) identifies the substitution elasticity and the bias of technology from wage variation across US plants, and [Zhang \(2019\)](#) estimates factor-augmenting productivity for capital, labor and materials in the Chinese steel industry. [Wang \(2025\)](#) uses cost minimization to estimate three factor-augmenting productivities in Chinese manufacturing and examines the effect of ownership change. [Demirer \(2026\)](#) generalizes the CES production function form with homothetic separability and labor-augmenting technology, and applies it to markup estimation. Our paper assumes the translog technology with labor-augmenting productivity and derives identification from cost minimization and a dynamic panel method, which is robust to product market power, following [Doraszelski and Jaumandreu \(2019\)](#) and [Jaumandreu \(2025\)](#). We apply the framework to examine the AI effect on factor-biased productivity in Korea.

Second, the paper contributes to the literature on endogenous productivity, in which the law of motion of productivity depends on firm states or choices. [Aw et al. \(2011\)](#) let R&D and the export status enter the productivity process jointly. [Doraszelski and Jaumandreu \(2013\)](#) model R&D as a determinant of future productivity, and [De Loecker \(2013\)](#) lets the export status shift the Markov process. [Braguinsky et al. \(2015\)](#) study productivity around ownership changes in Japanese cotton spinning, and [Chen et al. \(2021\)](#) apply the framework to Chinese privatization. [Doraszelski and Jaumandreu \(2018\)](#) and [Wang \(2025\)](#) extend the shifter beyond total factor productivity to the factor-augmenting processes. We let AI, a general-purpose technology, shift both labor-augmenting and Hicks-neutral productivity.

Third, the paper contributes to the firm-level evidence on artificial intelligence. Task experiments measure labor efficiency directly, since output per worker rises with access to an AI tool in writing and in customer service (Noy and Zhang, 2023; Brynjolfsson et al., 2025), and they hold fixed the two things a firm’s response depends on, its elasticity of substitution and the demand it faces. Czarnitzki et al. (2023) use a German innovation survey with an instrumental-variables design and find positive productivity effects, and McElheran et al. (2025) use US Census data on industrial AI and find J-curve returns. Where adoption is not surveyed, it has been proxied by patents (Da Silva Marioni et al., 2024) or by worker resumes (Babina et al., 2024). Park and Shin (2025) work with the survey we use and find that adoption raises labor productivity (measured by value added per worker). Each of these studies estimates one effect on output or on value added per worker, and one effect cannot separate the input the technology makes more efficient. We contribute firm-level evidence on AI augmenting labor, and a measure of how much of the adoption effect runs through labor efficiency. We also report how that effect differs across sectors and across the areas in which firms apply AI.

The remainder of the paper is organized as follows. Section 2 describes the Survey of Business Activities, the adoption and application-area measures built on it, and the production variables and cost shares that motivate the specification. Section 3 sets out the translog technology, the two productivity processes, and the identification and estimation. Section 4 reports the pooled estimates of the parameters of the production function and the two shifters of adoption. Section 5 splits the effect at the arrival of generative models, Section 6 across the sectors the estimator supports, and Section 7 across the four application areas. Section 8 takes the estimates to markups and to the aggregate. Section 9 concludes.

2 Background: AI Adoption in Korea

2.1 Survey of Business Activities

We use the annual Survey of Business Activities, administered by Statistics Korea, the national statistical office of South Korea. In all industries, the survey covers firms with at least 50 regular employees and at least 300 million Korean won in capital. These thresholds hold throughout the survey period.

Our baseline panel spans 2017–2024. The digital-technology questions begin in 2017, so the panel begins there, and the pre-adoption covariates are observed from the same year. The resulting panel holds 97,231 firm-year observations on 17,817 firms. We keep the twelve industry groups as the survey reports them, and the estimating equations absorb industry-by-year effects at that level.

2.2 AI-Related Variables

The survey records firm performance, diversification and group affiliation, performance-management systems, and shifts in strategy and industry structure ([Statistics Korea, 2022](#)). It has collected these characteristics since 2005. In 2017, it added a digital-technology module to track the diffusion of new technologies. Firms report whether they have adopted any of the nine technologies: artificial intelligence, robotics, the internet of things, cloud computing, big data, mobile technologies and services including 5G, blockchain, 3D printing, and augmented and virtual reality. They also report in which of the four application areas they use these technologies: product and service development, production processes, sales and marketing strategy, and organizational management.

AI Adoption Indicator [Statistics Korea \(2022\)](#) defines AI as the technology that underpins programs capable of human-like learning, reasoning, perception, understanding of

natural language, and related capabilities. Following the survey’s definition, we classify a firm as an adopter if it reports employing the technology in any of the four application areas. The survey counts use that the firm has adopted, encouraged or paid for at the corporate level, and not the informal use of an individual employee.²

This direct measure of use improves on common proxies in prior work, such as AI-related patents (Da Silva Marioni et al., 2024) or employee resumes (Babina et al., 2024). Those proxies capture development activity or hiring intent rather than deployment in production (Park and Shin, 2025). A patent count identifies a developer of the technology, and developing AI is a different thing from putting it to work in production.

AI Application Area Indicators Each adopter reports the area in which it applies AI: product and service development, production processes, sales and marketing strategy, or organizational management. The question asks for the firm’s main area, so the response is a single code and the four indicators are mutually exclusive within a firm-year.³

2.3 AI Adoption and Application

AI adoption in the panel as a whole rises from 1.3% in 2017 to 8.8% in 2024, and the pace varies across industries (Table 1). ICT leads, rising from 7.0% to 34.2%, and professional science and technology reaches 11.3% in 2024. Manufacturing, about half the panel, moves from 0.8% to 5.6%. Wholesale and retail reaches 7.3% and construction 5.8%, while transportation and warehouse at 2.9% and arts and sports at 2.4% remain the lowest. Electricity and gas records the second-highest rate in 2024 at 17.2%, on 64 firms, so that number rests on a small group.

Product and service development is the most commonly reported area, at 58.7% of

²Statistics Korea (2022) confirmed in correspondence that the question asks about AI in actual management activities, so an employee’s use of a tool such as ChatGPT counts once the firm adopts or pays for it.

³The area variable is recorded for every adopter firm-year and for no other, so the four indicators sum exactly to AI_{it} .

Table 1: AI adoption rate by industry in 2017 and 2024

Industry	2017		2024	
	AI-adopter share	Firms	AI-adopter share	Firms
Manufacturing	0.8%	5,927	5.6%	6,544
Electricity and gas	4.5%	44	17.2%	64
Construction	0.4%	467	5.8%	599
Wholesale and retail	0.9%	1,131	7.3%	1,635
Transportation and warehouse	0.2%	664	2.9%	716
Accommodation and food service	0.7%	303	4.1%	317
Information and communication	7.0%	967	34.2%	1,280
Real estate	0.0%	199	5.0%	242
Professional science and technology	1.3%	478	11.3%	626
Business facility management	1.3%	470	7.0%	632
Arts and sports	0.0%	261	2.4%	295
Other services	0.7%	292	7.5%	398
Total	1.3%	11,203	8.8%	13,348

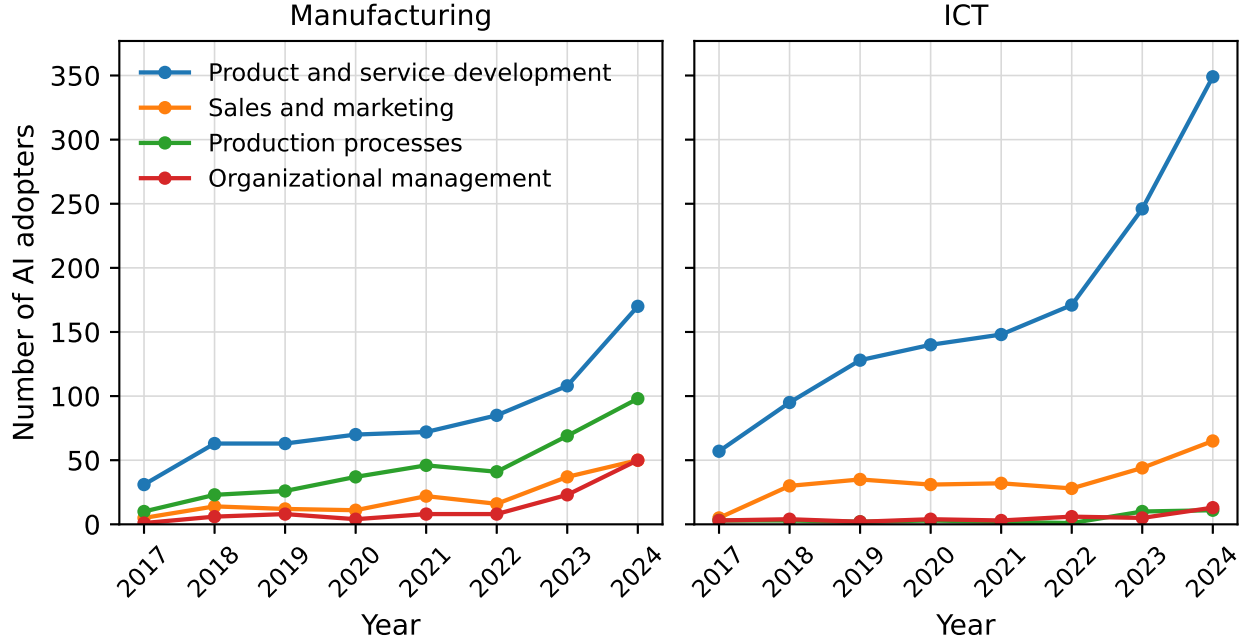
Notes: Other services pools agriculture, forestry and fishing, mining and quarrying, water supply, sewerage and waste management, education, human health and social work, and other service activities. Data source: Survey of Business Activities.

adopters in 2024, followed by sales and marketing at 18.6%. Organizational management and production processes each account for about an eighth. Figure 1 tracks the four areas in the two sectors the estimates support. Manufacturing spreads across the four, at 170 of 368 adopters in 2024 for product and service development and 98 for production processes, an area that rose from 10 in 2017 and is second in every year. In ICT, product and service development accounts for 349 of the 438 adopters and production processes for 11, so one area carries the sector. Production processes and organizational management stay in single digits in ICT through 2022, while in manufacturing both accelerate after that year, to 98 and 50 in 2024.

2.4 Production Variables

Table A.2 reports summary statistics for the production variables. The number of firms increases from 11,203 to 13,348 over the sample period. The average firm revenue fluctuates

Figure 1: AI application areas in manufacturing and in ICT (2017–2024)



Notes: Number of firms reporting each area as their main application of AI. Each adopter reports one area, so the counts do not overlap. Both panels plot counts. Data source: Survey of Business Activities.

between 173 and 189 billion KRW, without trend. The capital stock rises from 88 billion KRW in 2017 to 102 billion KRW in 2024.⁴ Labor input, measured in employees, stays between 337 and 356 workers per firm, and intermediate input between 132 billion KRW in 2018 and 143 billion KRW in 2021.⁵ We deflate revenue and intermediate input by the producer price index and capital by the gross fixed capital formation deflator, with 2017 as the base year. Firm-level prices are not collected, so the productivity we estimate is revenue-based.

The annual average input cost shares over 2017–2024 appear in Table A.3. Intermediate input is the largest cost component, at 68 to 69% of total cost throughout the period. Labor is second at an average of 26.0%, and capital smallest at about 5.5% in every year.

⁴Capital is a perpetual-inventory stock anchored on the firm's first observed book value.

⁵We use intermediate input rather than the survey's narrow materials line, which is zero for most firms outside manufacturing.

3 Empirical Framework

The technology is translog with two productivity states, one Hicks-neutral and one labor-augmenting. Each state evolves as a controlled Markov process shifted by AI adoption. Robot use enters the same way, as a comparison. Identification uses cost minimization over the variable inputs, which leaves the firm's conduct in the output market unrestricted, and estimation uses dynamic panel GMM.

3.1 Translog Production Function

We model firm i 's gross output in year t with a translog production function. Following [Doraszelski and Jaumandreu \(2019\)](#), we take it to be separable in capital and homogeneous of degree ν in labor and the intermediate input,

$$y_{it} = \alpha_0 + \alpha_K k_{it} + \alpha_L (\omega_{it}^L + \ell_{it}) + \alpha_M m_{it} - \frac{\alpha}{2} (m_{it} - \ell_{it} - \omega_{it}^L)^2 + \omega_{it}^H + \varepsilon_{it}, \quad (1)$$

where y_{it} is log deflated revenue and k_{it} , ℓ_{it} , and m_{it} are logs of deflated capital, employment, and deflated intermediate input. Homogeneity of degree $\nu \equiv \alpha_L + \alpha_M$ in the two variable inputs imposes $-\alpha_{LL} = -\alpha_{MM} = \alpha_{LM} \equiv \alpha$, which collapses the three second-order terms into the single quadratic in (1). Returns to scale are $\alpha_K + \nu$.

Labor-augmenting productivity ω_{it}^L enters multiplicatively with labor, so that $e^{\omega_{it}^L} L_{it}$ is labor measured in efficiency units, while ω_{it}^H shifts output whatever the input mix. The disturbance ε_{it} collects measurement error and shocks that cannot be anticipated when inputs are chosen.

The output elasticities of the two variable inputs are

$$\begin{aligned} \varepsilon_{L,it} &= \alpha_L + \alpha(m_{it} - \ell_{it} - \omega_{it}^L), & \varepsilon_{M,it} &= \alpha_M - \alpha(m_{it} - \ell_{it} - \omega_{it}^L), \\ \varepsilon_{L,it} + \varepsilon_{M,it} &= \nu, \end{aligned} \quad (2)$$

and the elasticity of substitution between them is

$$\sigma_{it} = \frac{\nu S_{it}^L (1 - S_{it}^L)}{\nu S_{it}^L (1 - S_{it}^L) + \alpha}, \quad (3)$$

where S_{it}^L is the labor share of the variable cost. Unlike a constant-elasticity-of-substitution production function, the translog lets the elasticity vary with the observed cost share.

3.2 Two Productivity Processes

Both productivity states follow a first-order controlled Markov process in which technology adoption shifts the law of motion:

$$\omega_{it}^H = \rho_H \omega_{i,t-1}^H + \gamma^H AI_{it} + \kappa^H Rob_{it} + \delta_{j(i)t}^H + \xi_{it}^H, \quad (4)$$

$$\omega_{it}^L = \rho_L \omega_{i,t-1}^L + \gamma^L AI_{it} + \kappa^L Rob_{it} + \delta_{j(i)t}^L + \xi_{it}^L. \quad (5)$$

The two processes are symmetric in their functional form, shifter, timing, and residual structure. [Doraszelski and Jaumandreu \(2018\)](#) specify them this way with R&D as the shifter, and [Wang \(2025\)](#) with ownership.

We carry a robot indicator in both processes because robots act on labor as well. A robot can raise what a given number of workers produce, and it can take over their tasks, so robot use can shift either state. Firms that adopt one technology are more likely to adopt the other, so leaving Rob_{it} out would load part of what robots do onto γ^L and γ^H . With it in, the two AI coefficients are the shifts at adopters holding robot use fixed, and κ^L is a comparison for γ^L .

We date both indicators at t . The indicator AI_{it} equals 1 if the firm i reports using AI for business purposes in year t , and 0 otherwise, and Rob_{it} equals 1 if it reports using robots. The same module of the survey asks about both, so the two are measured in the same way. We assume the firm decides whether to adopt at the start of t , before the innovations (ξ_{it}^H, ξ_{it}^L) are realized, so both indicators are predetermined with respect to them

and $\mathbb{E}[\xi_{it}^H \mid \omega_{i,t-1}^H, AI_{it}, Rob_{it}] = \mathbb{E}[\xi_{it}^L \mid \omega_{i,t-1}^L, AI_{it}, Rob_{it}] = 0$.⁶ The rest of the timing is standard. Capital is predetermined by prior investment, the firm learns $(\omega_{it}^H, \omega_{it}^L)$, chooses (ℓ_{it}, m_{it}) , and ε_{it} is realized last. As a robustness check, Appendix B reports the lagged specification, in which $AI_{i,t-1}$ and $Rob_{i,t-1}$ replace AI_{it} and Rob_{it} in the two processes and in both instrument sets.

The two coefficients of AI adoption are not on the same scale. A unit of ω^H is a unit of log output, but a unit of ω^L is a unit of log labor in efficiency units. Equation (1) converts it at the labor elasticity, $\partial y_{it} / \partial \omega_{it}^L = \varepsilon_{L,it}$. We therefore report the output-equivalent effect $\varepsilon_L \gamma^L$, evaluated at the sample mean labor share of the variable cost, and make all comparisons across specifications on that scale.⁷ The total effect of adoption on log output implied by (4)–(5) is $\gamma^H + \varepsilon_L \gamma^L$. Whether AI is labor-augmenting is then a question about $\varepsilon_L \gamma^L$. We ask whether it is positive and how it compares with γ^H . Persistence is estimated freely, and we report $\hat{\rho}_H$ and $\hat{\rho}_L$.

The terms $\delta_{j(i)t}^H$ and $\delta_{j(i)t}^L$ are effects specific to the firm i 's industry j in year t , where j indexes the twelve industry groups as the survey reports them. They absorb shocks common to a group in a given year. The output deflator varies at the level of the Korean Standard Industrial Classification section. Eleven of the twelve industry groups are a single section, so within those eleven the deflator is absorbed by $\delta_{j(i)t}^H$.

3.3 Identification and Estimation

Identification rests on cost minimization over the two variable inputs, not on profit maximization. The first-order conditions of a cost-minimizing firm involve the input prices, the technology and an unknown markup. They contain no elasticity of demand and no rule for how the firm sets its price (Doraszelski and Jaumandreu, 2019; Jaumandreu, 2025; Wang,

⁶Firms that report AI in one year do not always report it in the next, so $AI_{i,t-1}$ is a noisy proxy for use in year t . Chen et al. (2021) also date their privatization indicator at t in the law of motion for productivity, with the decision itself taken earlier.

⁷This is the output effect in Doraszelski and Jaumandreu (2018). It is a first-order quantity. Holding inputs fixed, a finite γ^L moves log output by $\varepsilon_L \gamma^L - (\alpha/2)(\gamma^L)^2$, and the second term is 0.02 percentage points at the pooled estimates and no more than 0.12 in any column we report.

2025). The firm takes input prices as given, as in the past literature (Doraszelski and Jaumandreu, 2018; Raval, 2019; Zhang, 2019; Demirer, 2026).⁸ Its conduct in the output market is unrestricted, so the markup $\mu_{it} = P_{it}/MC_{it}$ is an unknown that we do not model. For $X \in \{L, M\}$ the first-order condition is $P_{X,it}X_{it} = (P_{it}Y_{it}/\mu_{it})\varepsilon_{X,it}$, where $P_{L,it} \equiv W_{it}$ is the wage and $P_{M,it} \equiv P_{it}^M$ the price of the intermediate input, and $(w - p^M)_{it}$ is the log of their ratio.

Dividing one by the other eliminates the markup, output, and the output price:

$$\frac{\varepsilon_{L,it}}{\varepsilon_{M,it}} = \frac{W_{it}L_{it}}{P_{it}^M M_{it}} = \frac{S_{it}^L}{1 - S_{it}^L}. \quad (6)$$

Combined with (2) this gives $\varepsilon_{L,it} = \nu S_{it}^L$ exactly, and solving for the unobserved state inverts the first-order condition into observables (Doraszelski and Jaumandreu, 2019):

$$\omega_{it}^L = (m_{it} - \ell_{it}) + \frac{\alpha_L}{\alpha} - \theta S_{it}^L, \quad \theta \equiv \frac{\nu}{\alpha}. \quad (7)$$

The inversion contains $\theta = \nu/\alpha$ and not ν and α separately, and by (3) the entire path of σ_{it} depends only on θ .

Substituting (7) into (1) removes ω_{it}^L from the production function. The linear term in S_{it}^L cancels exactly and α_L is absorbed into the constant, leaving

$$y_{it} = \tilde{\alpha}_0 + \alpha_K k_{it} + \nu m_{it} - \frac{\nu\theta}{2} (S_{it}^L)^2 + \omega_{it}^H + \varepsilon_{it}. \quad (8)$$

Neither revenue nor the output price enters (7), so the labor-augmenting state is identified without the firm's own product price.

Quasi-differencing (7) with (5) and (8) with (4) yields two estimating equations, the first quasi-differenced at ρ_L and the second at ρ_H . Their residuals carry the productivity

⁸This assumption rules out labor market power.

innovations,

$$(m_{it} - \ell_{it}) - \rho_L(m_{i,t-1} - \ell_{i,t-1}) - \theta(S_{it}^L - \rho_L S_{i,t-1}^L) - \gamma^L AI_{it} - \kappa^L Rob_{it} - \delta_{j(i)t}^L = \xi_{it}^L, \quad (9)$$

$$(y_{it} - \rho_H y_{i,t-1}) - \alpha_K(k_{it} - \rho_H k_{i,t-1}) - \nu \left[(m_{it} - \rho_H m_{i,t-1}) - \frac{\theta}{2} ((S_{it}^L)^2 - \rho_H (S_{i,t-1}^L)^2) \right] - \gamma^H AI_{it} - \kappa^H Rob_{it} - \delta_{j(i)t}^H = \xi_{it}^H + \varepsilon_{it} - \rho_H \varepsilon_{i,t-1}, \quad (10)$$

where $\delta_{j(i)t}^L$ and $\delta_{j(i)t}^H$ are parameters free in each industry and year. They carry the industry-year effects of (4) and (5) together with the constants of (7) and (8). We refer to (9) as the share equation and to (10) as the output equation.

We estimate the two equations sequentially using the generalized method of moments (Hansen, 1982). The first step uses the share equation alone and identifies $(\theta, \rho_L, \gamma^L, \kappa^L)$. The second step imposes $\hat{\theta}$ in the output equation and identifies $(\alpha_K, \nu, \rho_H, \gamma^H, \kappa^H)$. Standard errors are clustered at the firm level and account for the first-step estimation of θ .⁹

The exclusion restrictions are on productivity innovations rather than on productivity itself, so a variable may be correlated with $\omega_{i,t-1}^H$ or $\omega_{i,t-1}^L$ and still be a valid instrument (Blundell and Bond, 2000; Akerberg et al., 2015; Doraszelski and Jaumandreu, 2018). Capital is predetermined, since k_{it} is set by investment decided at $t - 1$. The two technology indicators are predetermined as well. The two equations do not face the same endogenous regressor, S_{it}^L in the share equation and $(S_{it}^L)^2$ in the output equation, so we allow the instrument sets to differ. The exact IV set is reported in the notes to each table. Each instrument set also carries the industry-year indicators. Writing z_{it}^L and z_{it}^H for the instruments beyond them, the moment conditions are

$$\mathbb{E}[z_{it}^L \xi_{it}^L] = 0, \quad \mathbb{E}[z_{it}^H (\xi_{it}^H + \varepsilon_{it} - \rho_H \varepsilon_{i,t-1})] = 0, \quad (11)$$

⁹Doraszelski and Jaumandreu (2018) proceed sequentially in the same way, imposing the parameters estimated in the labor-augmenting process on the second equation and correcting the standard errors accordingly. The two equations can also be estimated jointly, which gains efficiency and leaves identification unchanged.

and in the baseline specification, the two vectors are

$$z_{it}^L = (\ell_{i,t-1}, \ell_{i,t-1}^2, m_{i,t-1}, m_{i,t-1}^2, S_{i,t-1}^L, (S_{i,t-1}^L)^2, AI_{it}, Rob_{it})',$$

$$z_{it}^H = (k_{it}, k_{it}^2, k_{i,t-1}, \ell_{i,t-1}, m_{i,t-1}, S_{i,t-1}^L, (S_{i,t-1}^L)^2, (w - p^M)_{i,t-1}, AI_{it}, Rob_{it})'.$$

The share equation carries eight instruments for four parameters and the output equation ten for five, so the two impose four and five overidentifying restrictions. The sector and application-area estimates run under sets of their own, so these counts are specific to the pooled specification.

4 Baseline Estimates

Table 2 reports the baseline estimates of the share equation (9) and the output equation (10) on the pooled sample of 78,442 firm-year observations from 15,829 firms during 2018–2024, under the specification of Section 3 with both technology indicators in the two productivity processes and in both instrument sets.

4.1 The Production Function Parameters

The share equation identifies $\theta = \nu/\alpha$ at $\hat{\theta} = 7.287$. It does not identify ν and α separately, so $\hat{\alpha} = \hat{\nu}/\hat{\theta} = 0.115$ is recovered once the output equation delivers $\hat{\nu}$. The elasticity of substitution is $\hat{\sigma} = 0.5930$ at the mean labor share of variable cost. Across firm-years its median is 0.53. Both lie well below one, so labor and the intermediate input are gross complements.

The output equation, with $\hat{\theta}$ imposed, gives a small and precisely estimated capital elasticity, $\hat{\alpha}_K = 0.0177$, and a degree of homogeneity in the two variable inputs of $\hat{\nu} = 0.8375$. Returns to scale $\hat{\alpha}_K + \hat{\nu}$ are 0.8552. These magnitudes are close to those of the study we follow. Doraszelski and Jaumandreu (2019) report capital elasticities between 0.015 and

Table 2: Translog production function with labor-augmenting productivity

<i>Panel A: Production function</i>	
$\theta = \nu/\alpha$	7.287*** (0.731)
α	0.1149*** (0.0128)
α_K	0.0177*** (0.0045)
ν	0.8375*** (0.0172)
Returns to scale $\alpha_K + \nu$	0.8552 (0.0151)
<i>Panel B: Productivity processes</i>	
ρ_L	0.9577*** (0.0013)
γ^L	0.0606*** (0.0119)
κ^L	0.0066 (0.0120)
ρ_H	0.9730*** (0.0020)
γ^H	0.0023 (0.0045)
κ^H	-0.0011 (0.0036)
<i>Panel C: Implied</i>	
Elasticity of substitution σ	0.5930 (0.0242)
Output elasticity of labor ε_L	0.2314
Labor-augmenting effect $\varepsilon_L \gamma^L$ (%)	1.402*** (0.276)
Total effect $\gamma^H + \varepsilon_L \gamma^L$ (%)	1.637*** (0.326)
Mean labor share of variable cost	0.276
Firms	15,829
Observations	78,442

Notes: Standard errors clustered at the firm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Both technology indicators enter both productivity processes and both instrument sets. Instruments in the share equation are ℓ_{t-1} , ℓ_{t-1}^2 , m_{t-1} , m_{t-1}^2 , S_{t-1}^L , $(S_{t-1}^L)^2$, AI_t and Rob_t . Instruments in the output equation are k_t , k_t^2 , k_{t-1} , ℓ_{t-1} , m_{t-1} , S_{t-1}^L , $(S_{t-1}^L)^2$, $(w - p^M)_{t-1}$, AI_t and Rob_t . Data source: Survey of Business Activities.

0.109 and ν between 0.848 and 0.987 in Spanish industries.¹⁰

4.2 AI Adoption and the Two Productivity States

AI adoption shifts the labor-augmenting process by $\hat{\gamma}^L = 0.0606$ and the Hicks-neutral process by $\hat{\gamma}^H = 0.0023$. Both states move in the same direction with adoption, and only the labor-augmenting one is estimated precisely. On the output-equivalent scale of Subsection 3.2, the labor-augmenting channel raises log output by $\varepsilon_L \hat{\gamma}^L = 1.402\%$ at the mean labor share of variable cost, and by 1.039% at the median. The total effect implied by (4) and (5) is $\hat{\gamma}^H + \varepsilon_L \hat{\gamma}^L = 1.637\%$. Of those 1.637 percentage points, the labor-augmenting channel carries 1.402, and the Hicks-neutral one is not distinguishable from zero. Adoption increases output at these estimates by making labor more efficient, not by increasing the productivity of every input alike.¹¹

The robot indicator shifts neither process. Its labor-augmenting coefficient is $\hat{\kappa}^L = 0.0066$ and its Hicks-neutral one -0.0011 . Robots are reported by 1,719 firm-years of the estimation sample against 3,555 for AI, and 619 firm-years report both.

Both processes are highly persistent, $\hat{\rho}_L = 0.9577$ and $\hat{\rho}_H = 0.9730$. The two states are persistent to a similar degree.¹²

5 Heterogeneity over Time

The adoption indicator pools seven years, and the technologies firms have in mind when they report AI use shifted over that span. Generative models became available to firms at

¹⁰Doraszelski and Jaumandreu (2019) also estimate the short-run elasticity of scale ν to be below unity in all ten of their industries, and treat the gap as the technological component that separates a markup from a price-to-average-variable-cost margin.

¹¹The overidentifying restrictions are rejected in both equations, at $J = 323.6$ on four restrictions in the share equation and $J = 225.8$ on five in the output equation. The statistic grows with the sample against any departure from the moment conditions, and ours has 78,442 firm-years. Across the instrument sets used in the studies we follow, $\hat{\gamma}^L$ lies between 0.060 and 0.063.

¹²At persistence this close to a unit root, the long-run effect $\gamma^L/(1 - \rho_L)$ is mechanically large and numerically unstable, and we report only the short-run semi-elasticity.

the end of 2022, with the release of ChatGPT in November. The survey year is the activity year, so 2023 and 2024 are the two years of the panel after that date. The adopters of those two years differ from the earlier ones. In the estimation sample the adoption rate rises from 3.8% in 2022 to 5.8% in 2023 and 8.8% in 2024, and firms reporting AI for the first time account for 40% of adopter firm-years in each of the two years against 22% in 2022. The share applying AI mainly to organizational management rises from 5.0% to 11.1%. If either the technology or the firms using it changed at that date, a single shifter estimated over 2018–2024 averages different things.

We therefore split the indicator three ways, by the period and by when the firm first reported adoption. AI_{it}^1 marks the years up to 2022 of firms that had adopted by 2022, AI_{it}^2 the years from 2023 of the same firms, and AI_{it}^3 the years from 2023 of firms reporting AI for the first time in 2023 or 2024. The three sum to AI_{it} , and they replace it in (4), in (5) and in both instrument sets. The first two cells cover the same firms on either side of the break, so they hold the cohort fixed. What they isolate is the calendar together with the length of time those firms have had AI in place. The second and third cover the same two years and isolate the adopters, firms that already had AI in place and firms that took it up after the release. The production function and the two persistence parameters stay common to the three cells. They are identified from the variation of the whole panel. Estimated on the two years after the break alone, the share equation returns no positive value of θ , so the model cannot be estimated period by period.

Table 3 reports the split. The labor-augmenting shifter is positive and significant in each of the three cells, $\hat{\gamma}^L = 0.0547$ before the break, 0.0805 for the same firms after it and 0.0520 for the new adopters. The production function does not move. $\hat{\theta}$ goes from 7.287 to 7.236. A labor-augmenting shift appears in every period and cohort the split can separate.

The Hicks-neutral shifter is 0.0035 before the break, 0.0161 for the same firms after it, and -0.0112 for the new adopters. In the two years after the break, the firms that already had AI and the firms that took it up then have total effects of 3.47% and 0.08%, and most

Table 3: The AI effect by period and by adoption cohort

	Adopted by 2022		First adopted
	2018–2022 (1)	2023–2024 (2)	2023–2024 (3)
<i>Panel A: Productivity processes</i>			
γ^L	0.0547*** (0.0160)	0.0805*** (0.0231)	0.0520** (0.0226)
γ^H	0.0035 (0.0057)	0.0161* (0.0096)	−0.0112 (0.0077)
<i>Panel B: Effects on output</i>			
Labor-augmenting effect $\varepsilon_L \gamma^L$ (%)	1.264*** (0.370)	1.857*** (0.535)	1.201** (0.521)
Total effect $\gamma^H + \varepsilon_L \gamma^L$ (%)	1.617*** (0.427)	3.469*** (0.676)	0.083 (0.586)
<i>Panel C: Common to the three cells</i>			
$\theta = \nu/\alpha$		7.236*** (0.728)	
α		0.1155*** (0.0130)	
α_K		0.0178*** (0.0045)	
ν		0.8357*** (0.0173)	
ρ_L		0.9576*** (0.0013)	
ρ_H		0.9730*** (0.0020)	
κ^L		0.0068 (0.0120)	
κ^H		−0.0011 (0.0036)	
Elasticity of substitution σ		0.5913 (0.0243)	
Adopter firm-years	1,800	835	920
Firms		15,829	
Observations		78,442	

Notes: One estimation on the pooled sample. The adoption indicator is split into three, and all three enter both productivity processes and both instrument sets. AI_t^1 covers the years up to 2022 of firms that had adopted by 2022, AI_t^2 the years from 2023 of those same firms, and AI_t^3 the years from 2023 of firms reporting AI for the first time. The robot indicator Rob_t enters both processes and both instrument sets as a level shifter, not split by period. Standard errors clustered at the firm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Instruments in the share equation are ℓ_{t-1} , ℓ_{t-1}^2 , m_{t-1} , m_{t-1}^2 , S_{t-1}^L , $(S_{t-1}^L)^2$, AI_t^1 , AI_t^2 , AI_t^3 and Rob_t . Instruments in the output equation are k_t , k_t^2 , k_{t-1} , ℓ_{t-1} , m_{t-1} , S_{t-1}^L , $(S_{t-1}^L)^2$, $(w - p^M)_{t-1}$, AI_t^1 , AI_t^2 , AI_t^3 and Rob_t . Data source: Survey of Business Activities.

of that distance sits in this term rather than in their labor-augmenting effects, which are 1.86% and 1.20%. The comparison holds those two years fixed, so it cannot separate who the new adopters are from how recently they adopted.

6 Heterogeneity across Sectors

Neither the elasticity of substitution nor the two shifters need be common across sectors, and pooling imposes that they are. Therefore, we re-estimate the specification separately within two sections of the Korean Standard Industrial Classification. Manufacturing accounts for about half the firms in the panel, and ICT has the highest adoption rate of the twelve industry groups. The remaining sections hold too few firms that ever switch to estimate the shifter with any precision.

Table 4 reports the two side by side. Each is estimated on its own subsample, so the industry-by-year absorber of Table 2 is constant within a column and reduces to a year effect. Robot use is reported by 1,302 of manufacturing’s 40,077 firm-years but by only 110 of ICT’s 7,225. That sample is too small to estimate a robot shifter with any precision, so the ICT column leaves the robot indicator out of both productivity processes and both instrument sets and otherwise runs the specification and instruments of the manufacturing column.¹³

The production function is nearly the same across the two. The degree of homogeneity in the variable inputs is $\hat{\nu} = 0.9223$ in manufacturing and 0.9293 in ICT, the returns to scale are 0.9350 and 0.9322, and both lie above the pooled 0.8552. Persistence is interior in all four processes, between 0.949 and 0.969.

The two sectors separate in the elasticity of substitution and in the shifter. Labor and intermediate input are gross complements in both and more so in ICT, $\hat{\sigma} = 0.2830$ against 0.4375. The labor-augmenting shifter is $\hat{\gamma}^L = 0.0549$ in manufacturing and 0.0267 in ICT, and both are distinguishable from zero. The Hicks-neutral shifter is indistinguishable from

¹³With the robot indicator, the ICT column gives $\hat{\kappa}^L = 0.1617$ on these 110 firm-years, against -0.0159 on manufacturing’s 1,302. On a sample this small the estimate is imprecise and likely overstated.

Table 4: The AI effect by sector

	Manufacturing	ICT
<i>Panel A: Production function</i>		
$\theta = \nu/\alpha$	4.974*** (0.771)	1.737** (0.847)
α	0.1854*** (0.0295)	0.5349** (0.2713)
α_K	0.0127*** (0.0044)	0.0029 (0.0120)
ν	0.9223*** (0.0154)	0.9293*** (0.0431)
Returns to scale $\alpha_K + \nu$	0.9350 (0.0128)	0.9322 (0.0344)
<i>Panel B: Productivity processes</i>		
ρ_L	0.9590*** (0.0017)	0.9557*** (0.0039)
γ^L	0.0549*** (0.0124)	0.0267** (0.0127)
κ^L	-0.0159 (0.0107)	
ρ_H	0.9690*** (0.0035)	0.9490*** (0.0078)
γ^H	0.0005 (0.0041)	-0.0036 (0.0066)
κ^H	-0.0004 (0.0030)	
<i>Panel C: Implied</i>		
Elasticity of substitution σ	0.4375 (0.0381)	0.2830 (0.0989)
Output elasticity of labor ε_L	0.1790	0.3245
Labor-augmenting effect $\varepsilon_L \gamma^L$ (%)	0.983*** (0.223)	0.865** (0.412)
Total effect $\gamma^H + \varepsilon_L \gamma^L$ (%)	1.034*** (0.358)	0.502 (0.564)
Mean labor share of variable cost	0.194	0.349
Firms	7,953	1,519
Observations	40,077	7,225

Notes: Each sector is estimated on its own subsample. Standard errors clustered at the firm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Instruments in the share equation are ℓ_{t-1} , ℓ_{t-1}^2 , m_{t-1} , m_{t-1}^2 , S_{t-1}^L , $(S_{t-1}^L)^2$, $(w - p^M)_{t-1}$, AI_t and Rob_t . Instruments in the output equation are k_{t-1} , k_t^2 , ℓ_{t-1} , m_{t-1} , S_{t-1}^L , AI_t and Rob_t . Robot use is reported in only 110 of the 7,225 ICT firm-years, 1.52%, so the ICT column omits Rob_t from both productivity processes and both instrument sets. Data source: Survey of Business Activities.

zero in both, as it is in the pooled sample. In both sectors, the identified part of the effect runs through labor efficiency.

The conversion to output closes most of that gap. Since $\varepsilon_L = \nu S^L$, a coefficient converts at the sector’s own labor share of variable cost, and that share averages 0.349 in ICT against 0.194 in manufacturing. The labor-augmenting effect on output is $\varepsilon_L \hat{\gamma}^L = 0.983\%$ in manufacturing and 0.865% in ICT. Manufacturing’s shifter is twice as large but applies to little more than half as much labor, so on the output scale the two sectors differ by only 0.12 percentage points. Reading the sector split off γ^L alone would overstate the distance between the two.

7 Heterogeneity by Area of Application

The survey asks each adopter for its main area of application, so the four area indicators are mutually exclusive and sum to AI_{it} . We replace the adoption indicator in (4) and (5) by the vector of four, and the same four replace AI_{it} in both instrument sets, so adoption remains its own instrument. The robot indicator stays beside the four in the pooled and manufacturing columns, and the ICT column leaves it out for the reason given in Section 6. Every adopter firm-year in the estimation sample carries an area, so the four indicators account for adoption exactly. We date them at t and take each one to be predetermined with respect to the two innovations, as Subsection 3.2 assumes for adoption itself. There are 3,555 such firm-years pooled. Table 5 reports the unrestricted estimates for the pooled sample and for the two sectors of Section 6.

Relative to the restricted estimates, the production function barely moves. In the pooled column, the labor-augmenting shifter differs across the four areas, and the Hicks-neutral one does not. Production processes carry $\hat{\gamma}_a^L = 0.1258$ and product and service development 0.0590, against the 0.0606 of Table 2 under the restriction that the four are equal. Sales and marketing at 0.0362 and organizational management at 0.0112 are indistinguishable

Table 5: The AI effect by area of application

	All sectors	Manufacturing	ICT
<i>Panel A: Labor-augmenting shifter γ_a^L</i>			
Product and service development	0.0590*** (0.0151)	0.0226 (0.0175)	0.0368*** (0.0142)
Sales and marketing	0.0362 (0.0270)	0.0662* (0.0352)	0.0005 (0.0280)
Production processes	0.1258*** (0.0264)	0.1065*** (0.0226)	-0.0342 (0.0395)
Organizational management	0.0112 (0.0320)	0.0579* (0.0305)	-0.0667 (0.0459)
<i>Panel B: Hicks-neutral shifter γ_a^H</i>			
Product and service development	0.0017 (0.0058)	0.0033 (0.0057)	-0.0039 (0.0074)
Sales and marketing	0.0046 (0.0095)	0.0012 (0.0137)	-0.0046 (0.0130)
Production processes	-0.0059 (0.0083)	-0.0005 (0.0059)	-0.0032 (0.0310)
Organizational management	0.0153 (0.0099)	-0.0110 (0.0087)	0.0190 (0.0182)
<i>Panel C: Labor-augmenting effect on output $\varepsilon_{L,a}\gamma_a^L$ (%)</i>			
Product and service development	1.510*** (0.387)	0.368 (0.286)	1.241*** (0.478)
Sales and marketing	0.813 (0.608)	1.105* (0.588)	0.014 (0.852)
Production processes	1.767*** (0.371)	1.509*** (0.321)	-1.068 (1.237)
Organizational management	0.211 (0.601)	0.868* (0.457)	-1.756 (1.208)
<i>Panel D: Production function and productivity processes</i>			
α	0.1135*** (0.0127)	0.1806*** (0.0285)	0.5310** (0.2678)
α_K	0.0178*** (0.0046)	0.0127*** (0.0045)	0.0037 (0.0120)
ν	0.8368*** (0.0172)	0.9217*** (0.0155)	0.9266*** (0.0437)
ρ_L	0.9577*** (0.0013)	0.9589*** (0.0017)	0.9561*** (0.0039)
ρ_H	0.9730*** (0.0019)	0.9690*** (0.0035)	0.9500*** (0.0078)
Elasticity of substitution σ	0.5957 (0.0241)	0.4438 (0.0380)	0.2840 (0.0985)
Firms	15,829	7,953	1,519
Observations	78,442	40,077	7,225

Notes: Standard errors clustered at the firm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The pooled column runs under the instruments of Table 2 and the two sector columns under their own columns of Table 4. The pooled and manufacturing columns carry the robot indicator in both productivity processes and both instrument sets, and its coefficient is insignificant in both. The ICT column carries no robot indicator, as in Table 4. Data source: Survey of Business Activities.

from zero. None of the twelve Hicks-neutral coefficients in the table reaches twice its own standard error.

The two sectors hold those areas in opposite proportions. Production processes are 28.2% of manufacturing’s adopter firm-years against 1.8% of ICT’s, and product and service development 50.2% against 79.3%. In manufacturing, production processes carry the largest shifter, $\hat{\gamma}_a^L = 0.1065$, twice the 0.0549 of Table 4. Product and service development is indistinguishable from zero there, at 0.0226. In ICT the ordering reverses. Product and service development carries 0.0368, while production processes and organizational management hold only 27 and 33 adopter firm-years, so those two cells are unmeasured rather than zero. At four fifths of the sector’s adopters, product and service development is what the restricted $\hat{\gamma}^L = 0.0267$ of Table 4 averages. Manufacturing supplies 326 of the 437 production process firm-years in the pooled column, and ICT 1,161 of the 2,216 product and service development ones. Production process in the pooled column is therefore largely manufacturing’s, and product and service development about half ICT’s.

Each shifter converts to output at its own rate. Equation (6) gives $\varepsilon_{L,it} = \nu S_{it}^L$ exactly, so an area’s coefficient converts at that area’s own labor share of variable cost. Adopters that apply AI to production processes have a mean share of 0.168, against 0.306 for those that apply it to product and service development. At their own shares the labor-augmenting effect on output is 1.767% for production processes and 1.510% for product and service development in the pooled column, against 1.402% under the restriction. In ICT, product and service development gives 1.241% against the restricted 0.865% of Table 4. The conversion nearly closes the gap. The two areas differ by a factor of two in γ_a^L and by a quarter of a percentage point on the output scale. The area split identifies the channel through which adoption reaches output. It does not rank the areas by the size of the gain.

8 Discussion

8.1 AI Adoption and the Markup

Table 6: AI adoption and the markup

	Pooled	Manufacturing	ICT
AI adoption	0.0010 (0.0027)	0.0058 (0.0036)	−0.0099* (0.0059)
Median markup	0.8964	0.9920	1.0036
Firms	17,817	8,840	1,728
Firm-years	97,231	49,108	8,958

Notes: Standard errors clustered at the firm level in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. The dependent variable is the log markup of (12). Every column includes firm effects and industry-by-year effects. The pooled and manufacturing columns also control for robot use. The median markup is $\mu_{it} = \hat{\nu} P_{it} Y_{it} / VC_{it}$ at the $\hat{\nu}$ of the same column, taken over that column’s firm-years. Data source: Survey of Business Activities.

Adding the two first-order conditions of Subsection 3.3 instead of dividing them, and using $\varepsilon_{L,it} + \varepsilon_{M,it} = \nu$, gives an accounting identity:

$$VC_{it} \equiv W_{it} L_{it} + P_{it}^M M_{it} = \frac{\nu}{\mu_{it}} P_{it} Y_{it} \quad \Longleftrightarrow \quad \mu_{it} = \nu \frac{P_{it} Y_{it}}{VC_{it}}. \quad (12)$$

The markup requires no demand system and no assumption on conduct (Jaumandreu, 2025).

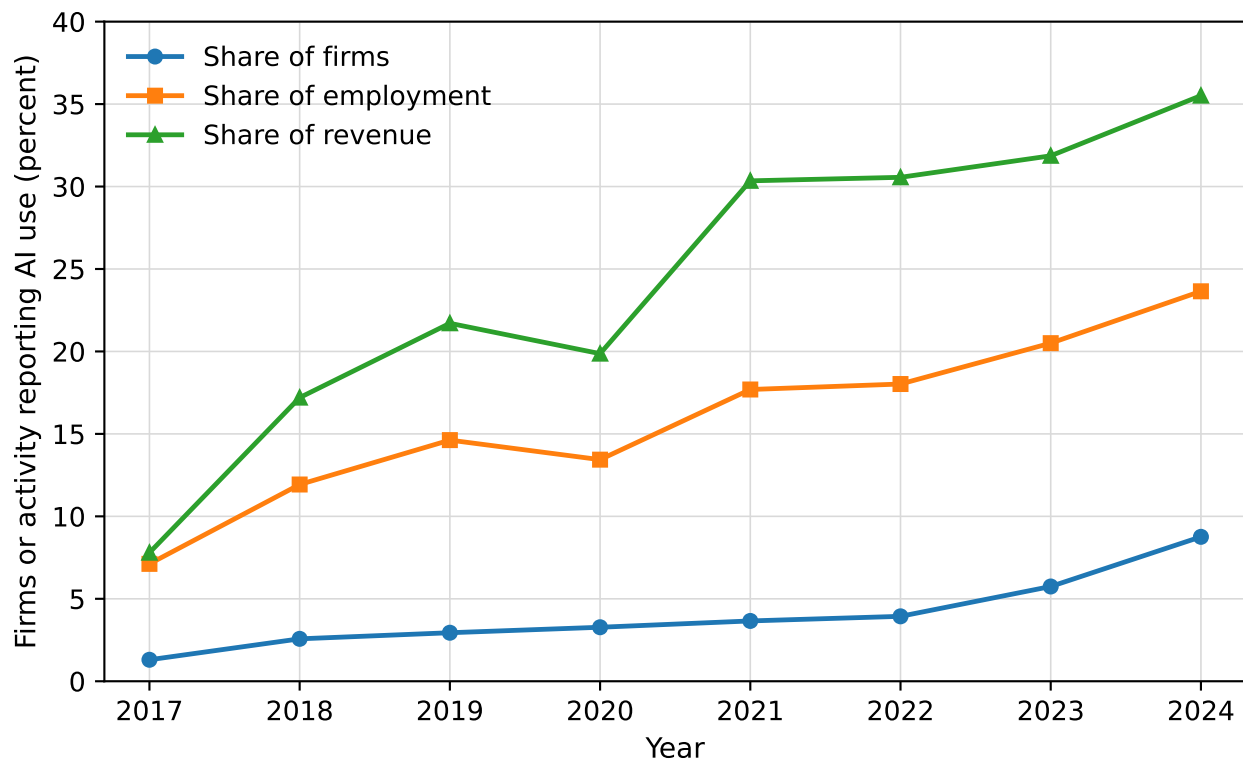
Measured output is deflated revenue, so ν is a revenue elasticity, and Bond et al. (2021) show that a markup built from a revenue elasticity has an estimand of one. The median markup in the table is below one in the pooled and manufacturing columns. It is $\hat{\nu}$ times the median ratio of revenue to variable cost, and that ratio exceeds one in all three columns, so the shortfall sits in $\hat{\nu}$. We therefore use (12) only for how the markup changes and not for its level. With ν common within a sample, the log markup and the log ratio of revenue to variable cost differ by a constant that firm effects absorb. We regress the log markup on the adoption indicator, as De Loecker and Warzynski (2012) do with export status. Firm effects

and industry-by-year effects are included throughout, and the robot indicator in the pooled and manufacturing samples. The pooled coefficient in Table 6 is 0.0010. The manufacturing coefficient is 0.0058 and not distinguishable from zero, and the ICT coefficient is -0.0099 , significant only at the 10% level. The coefficient is a within-firm association, not a causal effect. We do not find evidence that AI adoption raises the markup.

8.2 What the Firm-Level Effect Adds Up To

With each firm weighted by its share of the sample’s real revenue, s_{it}^Y , the contribution of adoption to measured output in year t is $\sum_i s_{it}^Y [\gamma^H + \nu S_{it}^L \gamma^L] AI_{it}$. In 2024 only 8.76% of firms report using AI, but those firms account for 35.51% of revenue and 23.65% of employment (see Figure 2).

Figure 2: AI adoption by firms, by employment and by revenue (2017–2024)



Notes: Each series is the share of that year’s total accounted for by firms reporting the use of AI, on the 97,231 firm-years that pass the panel’s restrictions. Data source: Survey of Business Activities.

Table 7 evaluates the contribution at the estimates of Table 2. Adoption increases 2024

Table 7: The aggregate contribution of AI adoption

	AI adoption rate (%)			Contribution to output (%)		
	Un-weighted	Revenue-weighted	ε_L at adopters	Hicks-neutral	Labor-augmenting	Total
2017	1.30	7.79	0.1115	0.018	0.053	0.071** (0.029)
2018	2.57	17.20	0.1096	0.040	0.114	0.155** (0.063)
2019	2.94	21.71	0.1105	0.051	0.145	0.196** (0.080)
2020	3.27	19.87	0.1130	0.047	0.136	0.183** (0.073)
2021	3.66	30.35	0.0925	0.071	0.170	0.241** (0.115)
2022	3.94	30.56	0.0916	0.072	0.170	0.241** (0.116)
2023	5.75	31.86	0.0982	0.075	0.190	0.264** (0.120)
2024	8.76	35.51	0.1063	0.083	0.229	0.312** (0.132)

Notes: Standard errors in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. The adoption rates are computed on the 97,231 firm-year panel, weighted by within-year shares of real revenue. The contribution columns use the estimates of Table 2. Standard errors are delta-method on $(\gamma^L, \nu, \gamma^H)$ with the weights held fixed. Data source: Survey of Business Activities.

sample output by 0.312%, of which 0.229 percentage points run through labor efficiency and 0.083 through the Hicks-neutral productivity. The production function is in gross output and the weights are shares of sales, so the figure is the change in the sample’s total gross output with every firm’s inputs held fixed. Only the current year’s shift in the two states enters the sum.¹⁴

Since $\varepsilon_{L,it} = \nu S_{it}^L$, the labor-augmenting part is the revenue-weighted adoption rate, 0.355, times the gain at adopters, 0.64% of their gross output. The gain is $\varepsilon_L \hat{\gamma}^L$ at the adopters’ labor share. At the sample mean the same product is the 1.402% of Table 2, more than twice as large. Adoption is concentrated in large firms, where labor is a small share of cost. The revenue-weighted labor share of variable cost at adopters is 0.127 in 2024, against a sample mean of 0.276. The labor share falls from 0.496 in the smallest revenue decile to 0.134 in the largest. That decile holds 77% of revenue and has the highest adoption rate at 12.9%.

[Acemoglu \(2025\)](#) bounds the gain from AI for the United States with the same two factors, the share of activity AI reaches and the saving on what it reaches. The tasks AI can profitably perform within ten years are 4.6% of GDP, and the cost saving on them is 14.4%, which gives 0.66% of TFP over the decade. Exposure in the survey is much wider than in his projection, and the gain per adopter much smaller.

9 Conclusion

This paper estimates a translog production function in which labor-augmenting and Hicks-neutral productivity follow their own Markov processes, each shifted by AI adoption. Cost minimization over the two variable inputs identifies both, so nothing is assumed about how the firm prices. The panel is Korean firms over 2017–2024.

Adoption raises labor-augmenting productivity by 6.1% and shifts the Hicks-neutral state

¹⁴The calculation is first-order accounting at the estimated parameters, not a general-equilibrium one. It holds inputs, prices, entry and exit fixed, attributes nothing to non-adopters, and covers the survey rather than the Korean economy.

by an amount indistinguishable from zero, so the 1.6% effect on output runs mainly through labor efficiency. Neither the sector split nor the application-area split moves the Hicks-neutral state, and in the period split only the cell for firms that already had AI in 2023–2024 carries a coefficient distinguishable from zero, at the 10% level. A robot indicator carried in the same two processes moves neither state.

The survey records whether a firm uses AI and nothing about the scale of that use. It reports no spending on the technology, and each adopter names one main application area, so breadth of use is invisible as well. Every coefficient here is the average shift at firms that report adoption, and what the gain is worth against what AI costs a firm lies outside these data. Section 5 finds a labor-augmenting shift in every period and cohort it can separate, which speaks to when a firm adopted and not to how much it uses.

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Appendices

A Data Appendix

A.1 Panel Construction

The raw extract of the Survey of Business Activities holds 107,075 firm-year records for 2017–2024. We keep incorporated firms with non-missing core accounts, and we harmonize industry codes over time. We follow the survey’s classification for multi-activity firms, assigning each to its primary industry and consolidating the sales of its secondary lines into that industry. Two features of the raw file are repaired first. One masked firm identifier carries 686 records with no revenue, no operating cost and zero employment. These records are the entire source of the file’s duplicated firm-year pairs, and we drop them. The mid-level industry label is recorded in English through 2023 and in Korean in 2024. We map one to the other using the 13,585 firms observed in both 2023 and 2024, which gives a one-to-one map on 65 labels.

Table A.1 reports the restrictions that define the panel. We require a positive value for output, capital, employment, intermediate input and the wage bill. We drop the financial and insurance section, where interest paid and insurance claims are a median 47% of reported operating cost and would enter a production measure as if they were purchased inputs. Three tail rules, all fixed before any estimate was inspected, keep firm-years whose ratio of variable cost to revenue lies in $[0.20, 2.00]$, whose ratio of capital to revenue lies in $[0.005, 20]$, and whose share of intermediate input in variable cost lies in $[0.02, 0.98]$. The first has a structural reading, since (12) makes that ratio ν/μ , but the interval is a data-quality screen and not a bound the model imposes.

The panel holds 97,231 firm-years on 17,817 firms. Both estimating equations are quasi-differenced and instrumented at $t - 1$, so estimation uses the 78,442 firm-years on 15,829 firms whose previous year is also observed.

A.2 Variable Definitions and Construction

Output and Its Price Revenue is total sales as reported by the firm, in current millions of Korean won. The price of output is the producer price index of the firm’s section of the Korean Standard Industrial Classification, published by Statistics Korea and rebased to 2017. It takes exactly one value within a section and year. Output is the ratio of the two, so a firm’s own price relative to its section index enters productivity.

Capital A perpetual-inventory stock, $K_{it} = (1 - \delta)K_{i,t-1} + (I_{it} - R_{it})/P_{it}^K$, where I_{it} and R_{it} are reported acquisitions and disposals of tangible and intangible assets, $\delta = 0.089$ is the median ratio of the depreciation charge to the lagged book stock, and P_{it}^K is the gross fixed capital formation deflator of the firm’s section, rebased to 2017. The anchor is the firm’s first observed book value of tangible assets plus non-negative intangible assets. Its half-life at $\delta = 0.089$ is 7.4 years, longer than the panel, and it still accounts for a median 0.85 of a firm-year’s stock. The measure is therefore the book stock reweighted by investment

Table A.1: Sample construction

	Removed	Firm-years	Firms
Firm-year records in the 2017–2024 extract		107,075	19,120
<i>Repairs to the raw file</i>			
Placeholder records with zero employment and no accounts	−686	106,389	19,119
<i>Sample restrictions</i>			
Non-positive output, capital, labor, intermediate input or wage bill	−271	106,118	19,063
Financial and insurance section	−2,941	103,177	18,600
Registered tail rules on VC/PY , K/PY and S^M	−5,946	97,231	17,817
Panel, 2017–2024		97,231	17,817
Observations with a one-year lag, 2018–2024		78,442	15,829

Notes: Restrictions are applied in the order shown, so each line reports what remains after the lines above it. The three tail rules keep firm-years with a ratio of variable cost to revenue in $[0.20, 2.00]$, a ratio of capital to revenue in $[0.005, 20]$, and a share of intermediate input in variable cost in $[0.02, 0.98]$. Evaluated one at a time on the non-financial sample, they remove 1,784, 3,474 and 1,251 firm-years, and they overlap. The last two lines differ because the estimating equations are quasi-differenced and instrumented at $t - 1$, which removes 2017. Data source: Survey of Business Activities.

flows rather than a stock whose starting value has decayed away. Land and construction in progress are a median 44% of book tangible assets in manufacturing and cannot be removed, since the asset-detail module is absent for 2017 and 2018.

Intermediate Input Operating cost net of the wage bill, the depreciation charge and advertising, divided by the price of output. The first three terms are the measure [Rhee and Pyo \(2010\)](#) use for Korean firm accounts, “cost of sales” plus “sales and general administration expenses” minus “personnel expenses” and “cost of depreciation”. Advertising is removed following [Doraszelski and Jaumandreu \(2019\)](#), who subtract it from intermediate consumption on the ground that it is not a production input. The survey reports operating cost both as a total and as its components, which for industries outside finance are materials cost, the wage bill, rent and lease payments, depreciation, taxes and dues, advertising and promotion, bad debt expense, and other operating expenses.ⁱ We call the variable intermediate input rather than materials, since it is not the survey’s materials-cost line, which is zero in a quarter of the panel, including 79% of transportation and warehouse firm-years and 49% of ICT against 1.3% in manufacturing.

Price of Intermediate Input The output price index of the firm’s section. The survey collects no separate input price index, so deflation is single.

ⁱThe survey records two further components, financial expenses and insurance expenses, only for the financial and insurance section, which we drop. Taxes and dues and bad debt expense are not production inputs either, but they are together 0.7% of what remains and we leave them in.

Employment Total number of workers at the end of the year, the sum of regular, temporary, and other workers as reported by the firm. Hours are not collected, and dispatched workers are excluded.

Wage Bill and Wage Total labor cost as reported by the firm, and that bill divided by employment. The survey does not separate production from non-production workers, so both are firm-wide.

Variable Cost and the Labor Share The variable cost is the wage bill plus the intermediate-input bill and S_{it}^L is the wage bill over that sum, with a mean of 0.277 in the 97,231 firm-year panel. Capital is outside variable cost, since the model takes first-order conditions for labor and intermediate input only and treats capital as predetermined.

A.3 Summary Statistics

Table A.2: Summary statistics by year: means

Year	No. of firms	Revenue	Capital	Labor	Intermediate input
2017	11,203	183,886.49	88,460.05	343.60	139,774.41
2018	11,739	174,300.14	87,471.60	336.67	131,673.29
2019	11,827	173,407.61	90,788.02	339.35	133,735.10
2020	11,909	178,110.07	94,003.72	345.89	136,309.59
2021	11,920	189,432.49	98,202.10	356.03	142,838.77
2022	12,268	185,085.15	97,769.03	354.30	141,827.92
2023	13,017	177,855.48	99,358.28	346.18	139,484.72
2024	13,348	180,566.80	101,522.26	350.16	137,768.08

Notes: For each year the table reports the number of firms in the panel and the means of four production variables: revenue, capital, labor, and intermediate input. Labor is the year-end headcount of employees. Intermediate input is non-labor variable cost, that is operating cost net of the wage bill, the depreciation charge and advertising. It is deliberately not called materials, because the survey's narrow materials line is zero for most firms outside manufacturing. Capital is a perpetual-inventory stock built from reported investment flows at a depreciation rate of 8.9%, anchored on the firm's first observed book value of tangible plus intangible fixed assets. All monetary values are in millions of Korean won (KRW 1M) and are deflated to a 2017 base. Revenue and intermediate input are deflated by the industry producer price index, capital by the investment price index. Data source: Survey of Business Activities.

Table A.3: Average annual input cost shares, 2017–2024

Year	Variable cost		Total cost		
	Labor	Intermediate input	Capital	Labor	Intermediate input
2017	0.269	0.731	0.054	0.253	0.693
2018	0.272	0.728	0.055	0.255	0.690
2019	0.275	0.725	0.056	0.258	0.685
2020	0.285	0.715	0.058	0.267	0.676
2021	0.280	0.720	0.056	0.262	0.681
2022	0.274	0.726	0.055	0.257	0.688
2023	0.275	0.725	0.054	0.259	0.687
2024	0.282	0.718	0.054	0.266	0.680
Mean	0.277	0.723	0.055	0.260	0.685

Notes: Shares are computed at the firm-year level and then averaged by year, and the Mean row is the average of the eight annual figures. The variable-cost columns divide the wage bill and the intermediate-input bill by their sum. These are the shares the estimating equations use, since the model takes first-order conditions for labor and intermediate input only. Capital is predetermined and its user cost is not observed. Intermediate input is non-labor variable cost, not the survey’s narrow materials line, which is zero for most firms outside manufacturing. The total-cost columns add a capital cost $uc \times K$, where K is a perpetual-inventory stock at a depreciation rate of 8.9% and $uc = 0.085$ is a Hall–Jorgenson user cost calibrated for this panel. That user cost is a descriptive convention rather than an estimated parameter, and it appears in no estimating equation. Widening it to $uc \in [0.06, 0.12]$ moves the mean capital share to 0.042 and 0.073 and the mean labor share to 0.264 and 0.255, leaving the time path unchanged. Data source: Survey of Business Activities.

A.4 Industry Correspondence

Table A.4: Industry correspondence

Survey industry group	KSIC section	Firm-years	Firms
Manufacturing	C	49,108	8,840
Electricity and gas	D	431	77
Construction	F	4,190	916
Wholesale and retail	G	11,018	2,280
Transportation and warehouse	H	5,409	942
Accommodation and food service	I	2,544	515
Information and communication	J	8,958	1,728
Real estate	L	1,779	390
Professional science and technology	M	4,444	926
Business facility management	N	4,269	905
Arts and sports	R	2,320	408
Other services	A+B+E+P+Q+S	2,761	608
Total		97,231	17,817

Notes: The twelve survey industry groups are the level at which the industry-by-year effects of equations (4) and (5) are absorbed. Eleven of them are a single section of the Korean Standard Industrial Classification. Other services pools six sections. The output deflator varies at the section level, so within that group it is not absorbed by the industry-by-year effect. Those six are A Agriculture, forestry and fishing, B Mining and quarrying, E Water supply, sewerage and waste management, P Education, Q Human health and social work, S Other service activities. Counts are on the panel. Firms are counted distinctly, so the Total is below the sum of the rows. Data source: Survey of Business Activities.

Industry codes follow the Korean Standard Industrial Classification, comparable to revision 4 of the International Standard Industrial Classification. The industry-by-year effects of (4) and (5) are absorbed in the twelve survey industry groups of Table A.4. Eleven of them correspond to a single section of the Korean Standard Industrial Classification. Other services pools six sections, so the output deflator is not absorbed within that group, 2.8% of the panel.

B The Timing of the Adoption Indicator

Both productivity processes of Subsection 3.2 carry AI_{it} and Rob_{it} and not their lags. The timing assumption stated there licenses that date. The firm decides whether to adopt at the start of t , before it learns the innovations (ξ_{it}^H, ξ_{it}^L) , so adoption is predetermined with respect to both. A one-year lag is the natural alternative, and it is the specification a reader who expects a delay between adoption and its effect on productivity would ask for.

Column (2) of Table B.1 replaces both indicators by their own one-year lags in the two Markov processes and in both instrument sets, so adoption stays its own instrument at the date it enters. The sample is unchanged at 78,442 firm-years on 15,829 firms, since both estimating equations are quasi-differenced and already require the previous year to be observed.

The production function does not move. $\hat{\theta}$ goes from 7.287 to 7.012 and the elasticity of substitution at the mean cost share from 0.5930 to 0.5837, $\hat{\nu}$ from 0.8375 to 0.8385 and $\hat{\alpha}_K$ from 0.0177 to 0.0174, none of them by as much as half its own standard error. Persistence is the same in both processes to three decimals.

The labor-augmenting shifter is slightly larger under lag, $\hat{\gamma}^L = 0.0631$ against 0.0606, a move of less than a fifth of its standard error, and the effect on output increases from 1.402% to 1.460%. The Hicks-neutral shifter rises from 0.0023 to 0.0071 and stays indistinguishable from zero, at $t = 1.34$. That movement is most of the change in the total effect, which goes from 1.637% to 2.168%. The channel through which adoption works does not depend on the date at which the indicator is read.

Table B.1: AI adoption dated at t and at $t - 1$

	(1) AI_{it}	(2) $AI_{i,t-1}$
<i>Panel A: Production function</i>		
$\theta = \nu/\alpha$	7.287*** (0.731)	7.012*** (0.712)
α	0.1149*** (0.0128)	0.1196*** (0.0135)
α_K	0.0177*** (0.0045)	0.0174*** (0.0044)
ν	0.8375*** (0.0172)	0.8385*** (0.0167)
Returns to scale $\alpha_K + \nu$	0.8552 (0.0151)	0.8559 (0.0147)
<i>Panel B: Productivity processes</i>		
ρ_L	0.9577*** (0.0013)	0.9576*** (0.0013)
γ^L	0.0606*** (0.0119)	0.0631*** (0.0135)
κ^L	0.0066 (0.0120)	0.0163 (0.0141)
ρ_H	0.9730*** (0.0020)	0.9730*** (0.0020)
γ^H	0.0023 (0.0045)	0.0071 (0.0053)
κ^H	-0.0011 (0.0036)	0.0038 (0.0049)
<i>Panel C: Implied</i>		
Elasticity of substitution σ	0.5930 (0.0242)	0.5837 (0.0247)
Output elasticity of labor ε_L	0.2314	0.2316
Labor-augmenting effect $\varepsilon_L \gamma^L$ (%)	1.402*** (0.276)	1.460*** (0.314)
Total effect $\gamma^H + \varepsilon_L \gamma^L$ (%)	1.637*** (0.326)	2.168*** (0.391)
Firms	15,829	15,829
Observations	78,442	78,442

Notes: Standard errors clustered at the firm level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) reproduces Table 2. Column (2) replaces the adoption and robot indicators by their one-year lags in the two productivity processes and in both instrument sets, so each remains its own instrument at the date it enters. The share equation is instrumented with ℓ_{t-1} , ℓ_{t-1}^2 , m_{t-1} , m_{t-1}^2 , S_{t-1}^L , $(S_{t-1}^L)^2$, and the output equation with k_t , k_t^2 , k_{t-1} , ℓ_{t-1} , m_{t-1} , S_{t-1}^L , $(S_{t-1}^L)^2$, $(w - p^M)_{t-1}$. Data source: Survey of Business Activities.